

Pairs of Consecutive and Near-Consecutive Powerful Numbers: Pell Families, Towers, and Localization

Berkay Yüksel Sayim
Independent Researcher, Germany
berksa@tutamail.com · ORCID 0009-0004-4993-7352

October 4, 2026

Abstract

We study the pairs $n, n + 1$ of powerful numbers, the pairs $n - 1, n + 1$ at distance two, and how their numbers grow. For distance one we establish $\liminf N_1(x)/\log x \geq 1.6822$, with a constant computed from Walker's families, list all pairs up to 2^{52} from our own enumeration, and show that the families containing a square alone outgrow a local probabilistic model, which nevertheless matches the counts in the computed range. For distance two we count the pairs in a box of kernels and heights as a sum over towers, derive an explicit lower bound at every height, and prove that the number of pairs at distance two up to x differs by $O(x^{2/9})$ from the number of kernels m with $m \mid U_1(m), T_1(m)$ even and $T_1(m) \leq x$, so that a bound x^θ for these kernels gives the bound $x^{\max(2/9, \theta)}$ for the pairs; under a squarefreeness assumption, the case in which Tao's argument for the bound $x^{2/5+o(1)}$ is weakest lies among these kernels. Every computed number is generated from the deposited data or from a cited constant, and every numbered statement is labeled as proved, taken from the literature, computed, measured, heuristic, or open.

About this series. This paper is one of three parts that share one source, one bibliography, and one numbering. A statement numbered I.4.5 carries that number in its part and in the collected volume. Every number that we computed is generated from the result files of the data deposit or from a cited constant; numbers from the literature carry a citation. The ranges and parameters of a computation and the constants of an argument are written in the text. The open questions of all parts are collected as Problems A to I in the framework introduction of the collected volume and in the file `OPEN_PROBLEMS.md` of the data deposit; both are generated from the parts. This part is a preprint. The author corrects any error that becomes visible after the release in a new version of the record; earlier versions remain available under their own version numbers.

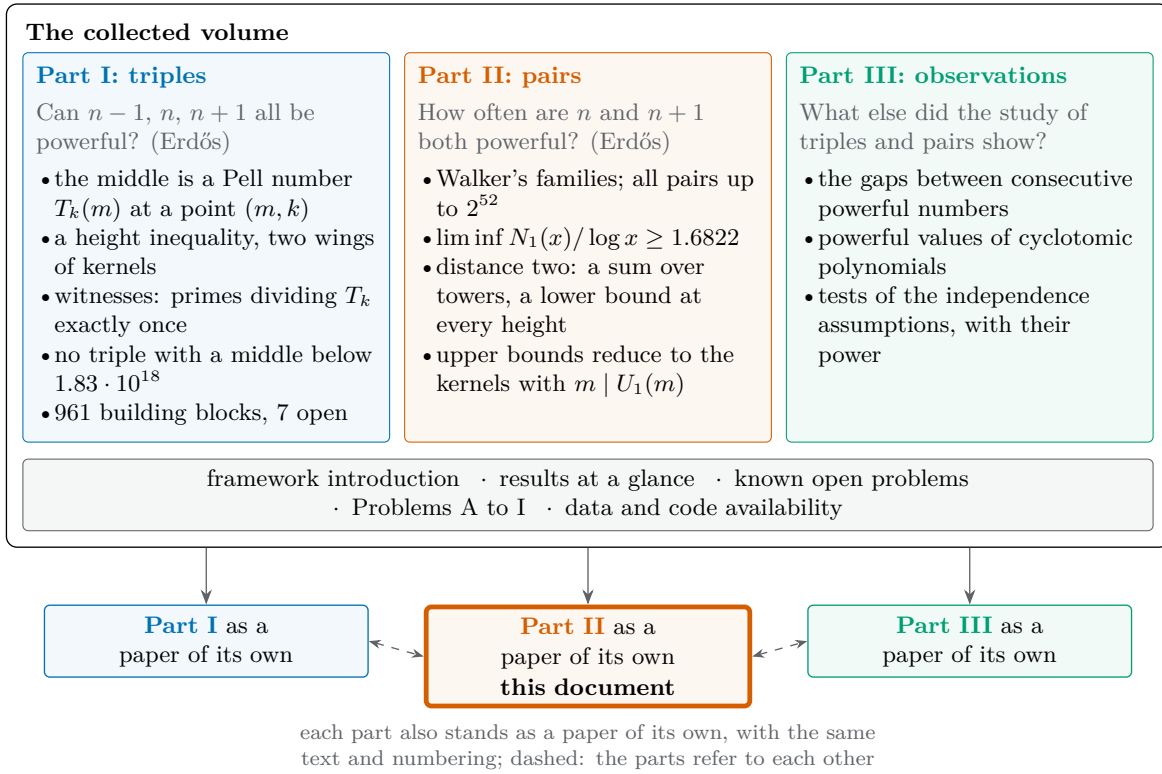


Figure 1: **What this research did.** It followed two questions of Erdős on consecutive powerful numbers, on triples and on pairs, through Pell equations, proved or computed what it could reach, and recorded the rest as observations and open problems. The collected volume contains the three parts and the framework that ties them together; each part also stands as a paper of its own, with the same text and numbering, and the parts refer to each other. The heavier frame marks the document in hand.

Contents

1	Introduction	4
2	Pell families	6
2.1	Pairs as solutions of one equation	6
2.2	The pairs inside one family	7
2.3	The families of the known pairs	7
2.4	Notation and terms at a glance	8
3	Enumeration	8
4	An explicit lower bound	10
5	A heuristic comparison model, and the convergence question	11
5.1	The comparison function	11
5.2	The families with a square overtake the comparison function	12

5.3	The convergence question	14
6	Pairs at distance two	15
6.1	The middles that are not divisible by four	15
6.2	The count is a sum over towers	16
6.3	Where the pairs live	18
6.4	How fast does the count grow?	20
7	Distance one along the Pell route	21
8	Open questions	22
	Data and code availability	23
	Use of AI tools	24

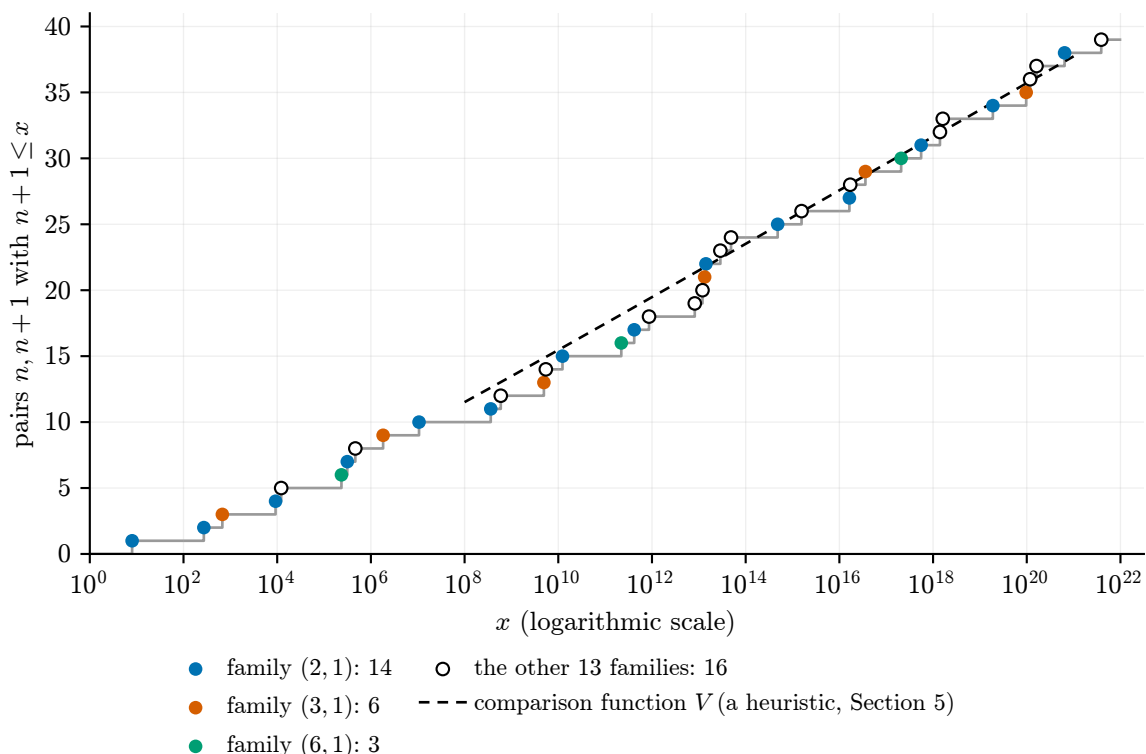


Figure 2: **The listed pairs up to n_0 , and the family each comes from.** The 39 pairs $n, n+1$ of powerful numbers with $n \leq n_0$ in the list of Proposition II.3.1(b), which is complete below 2^{52} , as a staircase: each step is one pair, and each point is colored by the family of Section 2 that contains it; the families with at least three pairs have their own color. The positions come from the family formulas of Proposition II.2.3, and at every power of ten from 10^8 to 10^{21} the staircase has exactly the number of pairs of our count, of which Table 3 shows every second one. The dashed line is the comparison function of Section 5, a heuristic yardstick, drawn from 10^8 on, where that table begins.

1 Introduction

A positive integer is *powerful* if every prime that divides it divides it at least twice. There are infinitely many n such that n and $n+1$ are both powerful; the solutions of Pell equations give them, as Walker recounts [20, p. 111]. Erdős asked two further questions [4]: whether all such pairs come from Pell equations, that is, whether n or $n+1$ must be a square, and whether the number $N_1(x)$ of such n with $n+1 \leq x$ is bounded by a power of $\log x$. The first question has a negative answer: Golomb’s pair 12167, 12168 has no square, and Walker showed that its family is infinite [20, p. 111]; Walker also described all pairs through Pell equations [20, Sections 2 and 3]. The second question is open. The best upper bound we know is $N_1(x) \ll x^{29/100+\epsilon}$, from a preprint of Reuss [13, Theorem 4], after $x^{2/5}(\log x)^2$ and $x^{7/19} \log x$ by Chan [5, Theorems 3 and 4] and x^γ for every $\gamma > 61/180$ by Blomer and Schöbel

[3, Theorem 3], all three for every fixed distance; Tao [16, Corollary 2.11] proves $x^{2/5+o(1)}$ for pairs at every fixed distance and discusses the question. The pairs at distance two, $n-1$ and $n+1$ powerful, are the objects of Part I, where their middles n carry the question of three consecutive powerful numbers; a question of Golomb about two consecutive odd powerful numbers was answered by Sentance [14, p. 272].

This part collects what we can prove, compute and measure about the count of pairs, at distance one and at distance two. It establishes an explicit lower bound of the order $\log x$, with a computed constant, checks the known list of pairs, and places the counting problem for distance two in the kernels that divide their own U_1 , of which only 3 with T_1 even have been found below $5.325 \cdot 10^{13}$, and which contain the case in which Tao's argument is weakest, under a squarefreeness assumption.

Main results.

- *An explicit lower bound* (Proposition II.4.1): $\liminf N_1(x)/\log x \geq 1.6822$, from the families of Walker with a square and squarefree part at most 10^6 , and from those without a square and $bd \leq 10^6$. That there are infinitely many pairs is classical; we did not find an explicit constant in the literature.
- *The count at distance two is a sum over towers* (Proposition II.6.2), and so $P_2(10^h) \geq 0.595h - 122$ for every h (Corollary II.6.3). In the box of kernels up to 10^8 and middles below 10^{2000} the list of pairs at distance two is complete: 371 with middle divisible by four (Part I) and 832 with middle $\equiv 2 \pmod{4}$ (Proposition II.6.1).
- *Localization* (Proposition II.6.4): up to $O(N^{2/9})$, the pairs at distance two with middle at most N are exactly the first pairs of the kernels m with $m \mid U_1(m)$, so every bound $E_3(N) \ll N^\theta$ on the number of these kernels gives $P_2(N) \ll N^{\max(2/9, \theta)}$. The case in which the argument of Tao is weakest lies among these kernels when the numbers n_2 and m_2 of Remark II.6.6 are squarefree.
- *The families with a square overtake the comparison function* (Proposition II.5.2): the count of pairs with a square in the families with squarefree part up to 10^4 exceeds the comparison function of Section 5 for every $x \geq e^{1000}$, and at every row of the table from 10^{130} on. So the agreement of such a model with the counts up to n_0 , which we also record, is a fact about that range and not a law.

How to read this part. Every statement is followed by a paragraph *In words* that says what it means with as few formulas as possible; a reader who wants the results can read the statements and these paragraphs. A reader who wants to check can read the proofs, the paragraphs *The computation*, and the tables; every number we computed is generated from the data files of the deposit or from a cited constant, and every numbered statement carries a label that says whether it is proved, taken from the literature, computed, measured, a heuristic, or open (see the table at the end). A statement that combines several kinds carries the label of the part that needs the most trust: a proof that uses our own computation is labeled computed, and a proof that uses a published result is labeled proved. Table 2 collects the notation.

Plan. Section 2 recalls Walker’s families and their periods. Section 3 lists the pairs up to 2^{52} from our own enumeration and checks the published list beyond it inside the families. Section 4 establishes the lower bound. Section 5 sets up the comparison function, marks it as a heuristic, shows that the square families overtake it, and states the convergence question. Section 6 treats the pairs at distance two: the second class of middles, the tower sum, the lower bound, the localization and the growth question. Section 7 puts the pairs at distance one on one Pell equation per kernel and tests the third neighbor. Section 8 collects the open questions.

2 Pell families

2.1 Pairs as solutions of one equation

As in Part I, $\text{sqfree}(N)$ is the product of the primes that divide N to an odd power, so that $N = \text{sqfree}(N) r^2$ for a positive integer r .

Lemma II.2.1 (powerful numbers). [proved] *A positive integer n is powerful if and only if $n = bY^2$ with $b = \text{sqfree}(n)$ and $b \mid Y$. This is the usual description of powerful numbers, in the form used below.*

Proof. Write $n = bY^2$ with $b = \text{sqfree}(n)$. For a prime $p \mid b$ we have $v_p(n) = 1 + 2v_p(Y)$, which is at least 2 exactly when $p \mid Y$. For a prime $p \mid n$ with $p \nmid b$ we have $p \mid Y$ and $v_p(n) = 2v_p(Y) \geq 2$ in any case. $\square$

In words. Every number is its squarefree part times a square. The number is powerful exactly when the squarefree part divides the root of the square part; for example $72 = 2 \cdot 6^2$ is powerful because $2 \mid 6$, and $12 = 3 \cdot 2^2$ is not because $3 \nmid 2$.

Proposition II.2.2 (Walker [20, Sections 2 and 3]). [literature] *Let n and $n + 1$ be powerful, and put $b = \text{sqfree}(n)$, $d = \text{sqfree}(n + 1)$, $Y^2 = n/b$, $X^2 = (n + 1)/d$. Then*

$$dX^2 - bY^2 = 1, \quad b \mid Y, \quad d \mid X.$$

Conversely, every solution in positive integers of this equation with b, d squarefree, $b \mid Y$ and $d \mid X$ comes from exactly one such pair, namely $n = bY^2$.

Proof. The first part is Lemma II.2.1 applied to n and to $n + 1$. Conversely, bY^2 and dX^2 are consecutive, and they are powerful by Lemma II.2.1, because $\text{sqfree}(bY^2) = b$ and $\text{sqfree}(dX^2) = d$ for squarefree b and d . The number n determines b, d, X and Y . $\square$

In words. A pair of consecutive powerful numbers is the same thing as a solution of $dX^2 - bY^2 = 1$ with two divisibility conditions. Golomb [8] sorted the pairs into two types, as Walker recounts in his introduction [20, p. 111], and Walker described both types through equations of this shape.

We call the set of pairs with a given (b, d) the *family* (b, d) . The order matters: in the family (b, d) the smaller number has squarefree part b . As $X^2 - Y^2 = 1$ has no solution in positive integers, b and d are not both 1. If $b = 1$ or $d = 1$, one of the two numbers is a square (Walker’s type I); if $b, d \geq 2$, neither is (type II). By Proposition II.2.2 every pair lies in exactly one family.

2.2 The pairs inside one family

For type I with $d = 1$ the equation is Pell's equation $X^2 - bY^2 = 1$. We use the notation of Part I with $m = b$: $\varepsilon_b = T_1 + U_1\sqrt{b}$ is its fundamental solution, $\varepsilon_b^k = T_k + U_k\sqrt{b}$, and $b' = \prod_{p|b, p \nmid U_1} p$.

Proposition II.2.3 (Walker [20, Theorems 2.2, 2.5, 3.2 and 3.5]). [literature]

- (i) Let $b \geq 2$. The pairs of the family $(b, 1)$ are $n = bU_k^2$, $n + 1 = T_k^2$ for exactly the $k \geq 1$ with $b' \mid k$.
- (ii) Let $d \geq 2$. The family $(1, d)$ needs a solution of $Y^2 - dX^2 = -1$. If there is none, or if d is even, the family is empty. Otherwise let $\eta = y_0 + x_0\sqrt{d}$ be the fundamental solution, $\eta^k = Y_k + X_k\sqrt{d}$ for odd k , and $g = d/\gcd(d, x_0)$. The pairs are $n = Y_k^2$, $n + 1 = dX_k^2$ for exactly the odd multiples k of g .
- (iii) Let $b, d \geq 2$, and let $\alpha = x_0\sqrt{d} + y_0\sqrt{b}$ be the smallest solution of $dX^2 - bY^2 = 1$ if there is one. All solutions are $\alpha^j = X\sqrt{d} + Y\sqrt{b}$ with j odd. If d is even and x_0 is odd, or b is even and y_0 is odd, the family is empty. Otherwise its pairs come from exactly the odd multiples j of ℓ , the product of the odd primes that divide bd but not x_0y_0 .

Proof. Part (i) is Lemma I.1.2 in [Part I] for $m = b$: $b \mid U_k$ if and only if $b' \mid k$. Part (ii) follows in the same way. For odd k the binomial theorem gives $X_k = \sum_{i \text{ odd}} \binom{k}{i} y_0^{k-i} x_0^i d^{(i-1)/2} \equiv k y_0^{k-1} x_0 \pmod{p}$ for a prime $p \mid d$, and $y_0^2 \equiv -1 \pmod{p}$, so y_0 is invertible modulo p . Hence $p \mid X_k$ if and only if $p \mid kx_0$. If d is even, then y_0 is odd, $dx_0^2 = y_0^2 + 1 \equiv 2 \pmod{8}$, so x_0 is odd and no X_k with k odd is even. If d is odd, $d \mid X_k$ if and only if $g \mid k$. Part (iii) is Walker's Theorems 3.2 and 3.5; the description of all solutions as odd powers of α is his Theorem 3.1, which he takes from [19, Theorem 9]. $\square$

In words. Inside one family the pairs come at regular steps. For the family $(b, 1)$ one takes every b' -th power of the Pell solution, for $(1, d)$ every $2g$ -th power starting at the g -th, and for type II the same with ℓ . We call b' , g and ℓ the *period* of the family. Walker proved all three parts; the proof of (i) and (ii) above is short enough to give here.

For the computations we used our own forms of these statements and checked them: part (i) against direct iteration modulo b for all 6082 squarefree $b \leq 10\,000$, and our residue class form of part (iii) against direct iteration modulo bd for all 955 solvable equations with $b, d \geq 2$ and $bd \leq 3000$; all agreed. For the families in Table 1 we also computed b' , g and ℓ from Walker's definitions and found the periods of our computation.

In words. The formulas were not only cited: a computer followed the powers one by one and found the pairs exactly where the formulas for parts (i) and (iii) say.

2.3 The families of the known pairs

Each family grows geometrically. Here and below, $\log$ is the natural logarithm; Table 1 gives $\log_{10}$ for readability. If n_1 is its first pair, the j -th is about $n_1 F^{j-1}$, where $\log F = 2b' \log \varepsilon_b$ for the family $(b, 1)$, $\log F = 4g \log \eta$ for $(1, d)$, and $\log F = 4\ell \log \alpha$ for type II. Throughout this part $n_0 = 3\,887\,785\,221\,910\,670\,811\,499$ is term number 39 of the list in Section 3, so $n_0 \approx 3.89 \cdot 10^{21}$. Table 1 lists the families of the 39 pairs of that list with $n \leq n_0$: 16 families,

32 pairs with a square and 7 without. The first pairs of the families (46, 1), (6, 1), (1, 5) and (3, 7) are the four examples in Walker's paper [20, pp. 113 and 116].

family (b, d)	period	$\log_{10} n_1$	$\log_{10} F$	pairs
(2, 1)	1	0.90	1.53	14
(3, 1)	3	2.83	3.43	6
(23, 2)	1	4.09	9.37	2
(6, 1)	3	5.37	5.97	3
(1, 5)	5	5.67	12.54	2
(46, 1)	1	8.77	9.37	2
(7, 2)	7	9.73	20.67	1
(5, 1)	5	11.94	12.54	1
(430, 1)	1	12.91	13.52	1
(79, 23)	1	13.08	27.36	1
(3, 7)	7	13.69	28.58	1
(10, 1)	5	15.19	15.79	1
(7, 1)	7	16.23	16.83	1
(14, 1)	7	20.07	20.67	1
(1022, 57)	1	20.21	41.62	1
(19, 35)	5	21.59	44.38	1

Table 1: The families of the pairs $n, n + 1$ of powerful numbers with $n \leq n_0 \approx 3.89 \cdot 10^{21}$, ordered by their first pair n_1 . The period is b' , g or ℓ of Proposition II.2.3; F is the growth factor from one pair of the family to the next. Each row is checked against three computations: the count in the list of pairs, the count predicted from the family, and, for type II, the pairs generated from the family and factored.

In words. Most of the known pairs sit in a few families. The family (2, 1), which starts with 8 and 9, alone has 14 of them, because its growth factor is small. Most other families contribute a single pair so far.

2.4 Notation and terms at a glance

Table 2 collects the symbols and words of this part, with the place where each is defined.

3 Enumeration

Proposition II.3.1 (the known pairs). [computed]

- (a) *There are exactly 26 pairs $n, n + 1$ of powerful numbers with $n + 1 \leq 2^{52}$.*
- (b) *Let n_0 be term number 39 of the b-file of OEIS A060355 [10], computed by D. Johnson, which lists such pairs by their smaller member n . The families of Section 2, with a square and squarefree part at most 10^6 or without a square and $bd \leq 10^6$, generate exactly 39 pairs with $n \leq n_0$ from their fundamental solutions, each checked by complete factorization and by the identity $dX^2 - bY^2 = 1$ with $b \mid Y$ and $d \mid X$. The first 26 of*

symbol or term	meaning	defined in
powerful	every prime factor occurs at least twice	Lemma II.2.1
$\text{sqfree}(N)$	the product of the primes that divide N to an odd power	§2
family (b, d)	the pairs $n, n+1$ with $\text{sqfree}(n) = b$ and $\text{sqfree}(n+1) = d$	§2
type I, type II	one of $n, n+1$ is a square ($b = 1$ or $d = 1$); neither is ($b, d \geq 2$)	§2
$\varepsilon_b, T_k, U_k, b'$	as in Part I, with the kernel $m = b$	Proposition II.2.3
η, g	the smallest solution $y_0 + x_0\sqrt{d}$ of $y^2 - dx^2 = -1$, and $g = d/\gcd(d, x_0)$	Proposition II.2.3
α, ℓ	the smallest solution $x_0\sqrt{d} + y_0\sqrt{b}$ of $dX^2 - bY^2 = 1$, and the product of the odd primes of bd that do not divide x_0y_0	Proposition II.2.3
period	b', g or ℓ : the step between the powers that give pairs	Proposition II.2.3
growth factor F	the factor from one pair of a family to the next; $\log F = 2b' \log \varepsilon_b, 4g \log \eta$ or $4\ell \log \alpha$	§2
n_0	term number 39 of the list of Section 3; the range of Table 1 is $n \leq n_0$	§2
$N_1(x)$	the number of pairs with $n+1 \leq x$	Proposition II.4.1
pair at distance two, middle	powerful numbers $n-1, n+1$; the number n between them	§6
kernel m, m' , lattice point	as in Part I: $m = \text{sqfree}(n^2 - 1)$, and $n = T_k(m)$ with k an odd multiple of m'	Lemma I.1.4 in [Part I]
$P_2(N), P_2(N; M)$	the number of middles $n \leq N$, all of them or those with kernel at most M	§6
discard criterion	a proved lower bound for $T_k(m)$ that removes a kernel before any T_k is computed	Theorem I.3.1 in [Part I]
tower	the middles $T_k(m)$ of one kernel m with $T_1(m)$ even, one for each odd multiple k of m'	§6
$w(m), c_M$	the rate $1/(2m' \log_{10} \varepsilon_m)$ of one tower in pairs per decimal digit, and the sum of these rates up to M	Proposition II.6.2
$E_J(N)$	the number of kernels with $m' < J$ and $T_1(m)$ even whose first middle is at most N	Proposition II.6.4
index m' , wing (ii), rung	the product of the primes of m that do not divide U_1 ; the kernels with $m \mid U_1$; one pair of a tower	Part I
$v_p(N), c_1, c_2, a_p$	the exponent of p in N ; the constants of $Q(x)$; the proportion of n with $v_p(n) = 1$	§5
$N_Q^*(L)$	the count of the pairs with a square from the family periods, by the floor formula	Proposition II.5.2
$Q(x), \rho(x)$	the number of powerful numbers up to x , and the derivative of its two main terms	§5
$C, V(x)$	the local correction factor and the comparison function, a heuristic yardstick	Definition II.5.1
$N_Q(x)$, born	the pairs with a square in the families with $b, d \leq 10^4$; a family is born at the logarithm of its first pair	Proposition II.5.2
D, R , third number	$D = \text{sqfree}(a(a+1))$, $a(a+1) = DR^2$; the neighbor $a-1$ or $a+2$ that is not excluded by 2	Proposition II.7.1

Table 2: Symbols and terms of Part II.

them are the pairs in (a), and the 39 pairs are exactly the first 39 terms of the b -file (Proposition II.3.2). That no pair with $n \leq n_0$ lies outside these families is a statement of that enumeration, not of ours.

The computation. (a) We listed all powerful numbers below 2^{52} block by block, the block k being $[2^k, 2^{k+1})$, each number written as a^2b^3 with b squarefree; this form is unique. The size of every block agrees with the count $\sum_b \lfloor \sqrt{x/b^3} \rfloor$ over squarefree b , taken at the two ends of the block. Inside a block the pairs are neighbors at distance one. A pair across two blocks would be $(2^k - 1, 2^k)$, and $2^k - 1$ is not powerful for $2 \leq k \leq 63$, by complete factorization. (b) The pairs are generated by the recursion of the Pell equation from the fundamental solution of each family, as in the proof of Proposition II.3.2; both members of every pair are factored completely, and the identity is checked with the squarefree parts b and d . The comparison with the b -file is made at run time against a copy that the reader supplies; the terms of the b -file are not stored in the data deposit. $\square$

In words. Up to 2^{52} we found all pairs ourselves. Beyond that we used a published list and checked that each of its entries really is a pair; we did not check that the list misses nothing.

Mian and Siddique [9] describe the larger enumerations as exhaustive below 10^{22} but not certified. We rely on it only where we say so. The next statement checks it inside the families of Section 2, independently of the list.

Proposition II.3.2 (completeness inside the families). [computed]

- (a) *In the families with a square, $(b, 1)$ and $(1, d)$ with $b, d \leq 10^4$, the pairs with $n \leq n_0$ are exactly the 32 pairs with a square among the terms of Proposition II.3.1(b). They lie in 10 families.*
- (b) *In the families without a square with $bd \leq 10^6$, the pairs with $n \leq n_0$ are exactly the 7 pairs without a square among those terms.*

The computation. For every family in the range we generated its pairs from Proposition II.2.3 up to n_0 and compared them with the terms, in both directions: no generated pair is missing from the terms, and no term with a square (in (a)) or without a square (in (b)) is missing from the generated pairs. For (b) each generated pair was also factored completely, and both numbers are powerful. $\square$

In words. Every one of the 39 known pairs comes out of our own family computation, and within the families we searched our computation finds no pair that the list lacks. So the list is complete for these families. Outside them, for example a family $(b, 1)$ with $b > 10^4$ whose first pair lies below n_0 , a missing pair would not be noticed here.

4 An explicit lower bound

Proposition II.4.1 (explicit lower bound). [computed] *Let $N_1(x)$ count the pairs $n, n+1$ of powerful numbers with $n+1 \leq x$. Then*

$$\liminf_{x \rightarrow \infty} \frac{N_1(x)}{\log x} \geq 1.6822.$$

The 676 094 nonempty families with a square and squarefree part at most 10^6 contribute 1.3685, and the 275 371 nonempty families without a square and with $bd \leq 10^6$ contribute 0.3137.

Proof. Let F be the growth factor of a nonempty family, as in Section 2. We first show that its j -th pair satisfies $n_j + 1 \leq F^j$. In the family $(b, 1)$ the j -th pair has $n_j + 1 = T_{jb'}^2$ by Proposition II.2.3(i), and $T_k = (\varepsilon_b^k + \varepsilon_b^{-k})/2 \leq \varepsilon_b^k$, so $n_j + 1 \leq \varepsilon_b^{2jb'} = F^j$. In the family $(1, d)$ the j -th pair comes from $k = (2j - 1)g$, and for odd k the conjugate of η^k is $-\eta^{-k}$, so $X_k\sqrt{d} = (\eta^k + \eta^{-k})/2 \leq \eta^k$ and $n_j + 1 = dX_k^2 \leq \eta^{2(2j-1)g} \leq \eta^{4jg} = F^j$. For type II the j -th pair comes from the exponent $(2j - 1)\ell$, the conjugate of $\alpha^i = X\sqrt{d} + Y\sqrt{b}$ is $X\sqrt{d} - Y\sqrt{b} = \alpha^{-i}$, and the same computation gives $n_j + 1 = dX^2 \leq \alpha^{4j\ell} = F^j$.

Hence a family has at least $\lfloor \log x / \log F \rfloor \geq \log x / \log F - 1$ pairs with $n + 1 \leq x$. Different families have different pairs by Proposition II.2.2. So for every finite set S of nonempty families

$$N_1(x) \geq \sum_{f \in S} \left(\frac{\log x}{\log F_f} - 1 \right), \quad \liminf_{x \rightarrow \infty} \frac{N_1(x)}{\log x} \geq \sum_{f \in S} \frac{1}{\log F_f}.$$

For the two sets named in the statement we computed the sums; the total is rounded down. $\square$

In words. Each family on its own produces pairs at a steady rate on a logarithmic scale: the number of its pairs below x grows like $\log x$ divided by the logarithm of its growth factor. Since no pair belongs to two families, the rates add up. Adding them for all families up to the stated sizes gives the constant. Families outside these ranges can only increase it.

5 A heuristic comparison model, and the convergence question

This section compares the count $N_1(x)$ of Section 4 with a function that a simple probabilistic picture suggests. The picture is a heuristic. We use the function only as a yardstick, and we do not claim that it predicts anything; the only statement of this section that is not a heuristic is Proposition II.5.2, and the lower bound of Section 4 does not depend on the picture.

5.1 The comparison function

Let $Q(x)$ be the number of powerful numbers up to x . Erdős and Szekeres found the main term, and Bateman and Grosswald proved $Q(x) = c_1\sqrt{x} + c_2x^{1/3} + O(x^{1/6})$ [6], [2, Theorem 1], with $c_1 = \zeta(3/2)/\zeta(3)$ and $c_2 = \zeta(2/3)/\zeta(2)$, so that $c_2 < 0$. Let $\rho(x) = c_1/(2\sqrt{x}) + c_2/(3x^{2/3})$ be the derivative of the two main terms.

Definition II.5.1 (comparison function; a heuristic). [heuristic] For a prime p let $a_p = (p - 1)/p^2$, the proportion of integers n with $v_p(n) = 1$, and let

$$C = \prod_p \frac{1 - 2a_p}{(1 - a_p)^2}, \quad V(x) = C \int_8^x \rho(t)^2 dt.$$

The picture behind V is this. If “ n is powerful” and “ $n+1$ is powerful” were independent events of probability $\rho(n)$ each, the expected number of pairs with $n+1 \leq x$ would be about $\int_8^x \rho(t)^2 dt$, which grows like $(c_1^2/4) \log x = 1.1808 \log x$. The events are not independent: a prime p cannot divide both n and $n+1$, so the events $v_p(n) = 1$ and $v_p(n+1) = 1$ exclude each other. Under the picture, the proportion of n with $v_p(n) \neq 1$ and $v_p(n+1) \neq 1$ is $1 - 2a_p$ instead of $(1 - a_p)^2$, and the factor C collects this correction over all primes, in the manner of the Hardy–Littlewood constants. Two assumptions enter here and are not derived: that the primes act independently, and that these proportions, which are densities over all integers, may be applied to powerful numbers; among powerful numbers the share of n with $4 \mid n$ has a different limit (Remark I.5.5 in [Part I]). Numerically $C = 0.7482$, so $V(x)$ grows like $0.8835 \log x$. Heuristics with a local factor at each prime are used for related counting problems, for example in [18].

In words. Powerful numbers get rarer as they grow, at a known rate. If two neighbors were powerful independently of each other, one could compute how many pairs to expect. They are not independent, because a prime never divides two neighbors at once, and C is one simple correction for that, not a derived one. The result is a curve to compare with, not a prediction.

Table 3 sets V against the count. Up to n_0 the list has 39 pairs; it is complete below 2^{52} and inside the families of Proposition II.3.2, and beyond that it is described as exhaustive but not certified [9]. The comparison function gives 38.94, and the independent picture gives 52.04. The constant C and the function ρ contain nothing fitted to the data: we computed V before we compared it with the terms beyond 2^{52} , where the count is 13 against 12.06 for V and 16.12 for the independent picture. The agreement is a fact about this range, within a scatter of about $\sqrt{N}$ for a count of N pairs; beyond 2^{52} the comparison does not separate V from the independent picture. The next statement shows that the agreement does not last.

$\log_{10} x$	pairs counted	$V(x)$	independent picture
8	10	11.5	15.4
10	14	15.5	20.7
12	18	19.5	26.0
14	24	23.5	31.4
16	26	27.6	36.9
18	31	31.6	42.3
20	35	35.7	47.7

Table 3: The number of pairs $n, n+1$ with $n+1 \leq x$, the comparison function $V(x)$ of Definition II.5.1, and the uncorrected integral $\int_8^x \rho^2$. The counts up to 2^{52} are our own enumeration; beyond it they come from the terms of Proposition II.3.1(b).

5.2 The families with a square overtake the comparison function

Let $N_Q(x)$ be the number of pairs $n, n+1$ with $n+1 \leq x$ that lie in a family $(b, 1)$ or $(1, d)$ with $b, d \leq 10^4$. By Proposition II.2.3 these families are known exactly. For a family f with first pair n_1 and growth factor F put $t_f = \log n_1$; we say the family is *born* at $\log x = t_f$. By the formulas of Section 2, n_j+1 is $(\alpha^k + \alpha^{-k})^2/4$ times a constant, with α the unit of the family and k growing by the period from one pair to the next; so $\log(n_j+1) = \log(n_1+1) + (j-1) \log F - \delta_j$

with $0 \leq \delta_j < 2 \log 2$, the correction coming from the decreasing terms α^{-k} . The count

$$N_Q^*(L) = \sum_f \max\left(0, \left\lfloor \frac{L - t_f}{\log F_f} \right\rfloor + 1\right),$$

over the families with $b, d \leq 10^4$, therefore agrees with $N_Q(e^L)$ except that a pair within $1/n_1$ of the boundary on the logarithmic scale may fall on the wrong side.

Proposition II.5.2 (the families with a square overtake the comparison function). [computed]

(a) Table 4 gives $N_Q^*(L)$ and $V(e^L)$ for 8 values of $L = \log x$ up to 1000. On a finer grid than the table, N_Q^* first exceeds V at $L = 192.9$, that is, at $x \approx 10^{83.8}$; at the next row of the table it is below V again, and from $L = 300$ on it is above V at every row of the table.

(b) For every $L \geq 1000$, $N_Q(e^L) > V(e^L)$.

$L = \log x$	$N_Q^*(L)$	$V(e^L)$	independent picture	rate of the families born
49.7	32	38.9	52.0	0.7049
75.0	54	61.3	81.9	0.7744
100.0	76	83.4	111.4	0.8136
150.0	124	127.5	170.4	0.8755
200.0	171	171.7	229.5	0.9011
300.0	265	260.1	347.6	0.9300
500.0	468	436.8	583.7	0.9765
1000.0	997	878.5	1174.1	1.0345

Table 4: The pairs with a square in the families with $b, d \leq 10^4$, counted exactly from the family periods, against the comparison function. The last column is $\sum 1/\log F$ over the families born by L , the rate at which N_Q grows there.

Proof. (a) is the computation of the table. (b) Let $\mathcal{F}$ be the set of the 103 families with $b, d \leq 10^4$ that are born by $L = 1000$, and let $s = \sum_{f \in \mathcal{F}} 1/\log F_f = 1.0345$. Write $y_f(L) = (L - t_f)/\log F_f$. The j -th pair of f has $\log(n_j + 1) \leq L$ as soon as $\log(n_1 + 1) + (j - 1)\log F_f \leq L$, that is, $j \leq (L - \log(n_1 + 1))/\log F_f + 1$; so f has more than $(L - \log(n_1 + 1))/\log F_f \geq y_f(L) - 1/(n_1 \log F_f)$ pairs with $n + 1 \leq e^L$, and the last term is below 0.05, because $n_1 \geq 8$ and $\log F_f \geq 2 \log(2 + \sqrt{3}) > 2.6$ for every family. On the other hand, each term of $N_Q^*(1000)$ is at most $y_f(1000) + 1$. So for $L \geq 1000$,

$$N_Q(e^L) > \sum_{f \in \mathcal{F}} y_f(L) - 0.05 |\mathcal{F}| \geq (N_Q^*(1000) - |\mathcal{F}|) - 0.05 |\mathcal{F}| + s(L - 1000)$$

and $0.05 |\mathcal{F}| < 6$, so

$$N_Q(e^L) > 894 - 6 + s(L - 1000).$$

On the other side, $\frac{d}{dL} V(e^L) = C \rho(e^L)^2 e^L = C(c_1/2 + (c_2/3)e^{-L/6})^2 < C c_1^2/4 = 0.8835$, because $c_2 < 0$. Hence $V(e^L) \leq V(e^{1000}) + 0.8835(L - 1000)$ with $V(e^{1000}) = 878.5$. Since $894 - 6 > 878.5$ and $s > 0.8835$, the claim follows. The value of $V(e^{1000})$ is rounded to one decimal and the product C is truncated; truncating C increases V , and the margin is larger than the rounding, so both errors are on the safe side. $\square$

In words. The comparison curve grows at a fixed rate on a logarithmic scale. The families with a square alone grow faster once enough of them have been born, and from 10^{130} on, at every row of the table, they produce more pairs than the curve gives for all pairs together; for all $x \geq e^{1000}$ this is proved. So the curve matches the counts only in the range we can see, and it must fail beyond it. It is a yardstick for the present, not a law.

5.3 The convergence question

By Proposition II.4.1, $\liminf N_1(x)/\log x$ is at least the sum of $1/\log F$ over any finite set of families. The sum over the families with a square and squarefree part at most B is 0.6055 for $B = 10$ and 0.8949 for $B = 100$; Table 5 continues it and adds the families without a square with $bd \leq B$.

$\log_{10} B$	with a square, squarefree part $\leq B$	without a square, $bd \leq B$	sum
3	1.1044	0.1783	1.2827
4	1.2235	0.2385	1.4619
5	1.3110	0.2831	1.5941
6	1.3685	0.3137	1.6822

Table 5: The partial sums $\sum 1/\log F$ over the families of Section 2. The last entry of the last column is the constant of Proposition II.4.1.

Question II.5.3 (convergence of the rates). [open] Does $\sum_f 1/\log F_f$, taken over all families, converge?

The increase per decade of B shrinks in both columns of the table. We counted where the last increase comes from: in the decade $10^5 < B \leq 10^6$, families with $\log \varepsilon \leq 100$ supply 79 % of it among the families with a square and 77 % among those without, and the largest one percent of the families supply 71 % and 72 %. So in the last decade we computed, the increase is carried by a small share of the families, those with small units, not by many small contributions. Whether this continues is not known.

A positive answer would not by itself give an asymptotic formula for $N_1(x)$. Families keep being born, and each newborn family adds a pair at once; turning the sum of the rates into the rate of $N_1(x)$ needs control of how many families are born below every height, which we do not have. The towers of Section 6 pose the same question for distance two, as Question II.6.7.

This is also where an answer to the second question of Erdős could come from. Every pair belongs to exactly one family, and a family has about $\log x / \log F$ pairs below x . So $N_1(x)$ is governed by two quantities: how many families are born below x , and the sum of their rates, which is Question II.5.3. If the number of families born below x were bounded by a power of $\log x$, so would be $N_1(x)$, and the answer to the question of Erdős would be yes. We know no such bound, and no bound for the number of families better than the upper bounds for $N_1(x)$ itself recalled in the introduction. We do not know whether it holds.

In words. Every family contributes a rate, and the rates add up. Whether the total of all rates is finite is open. Our partial sums grow more and more slowly, but they are driven by a small share of the families, those with small units, so a few more decades would not settle it. Erdős's question would be answered by showing that new families are born rarely enough; we cannot show this.

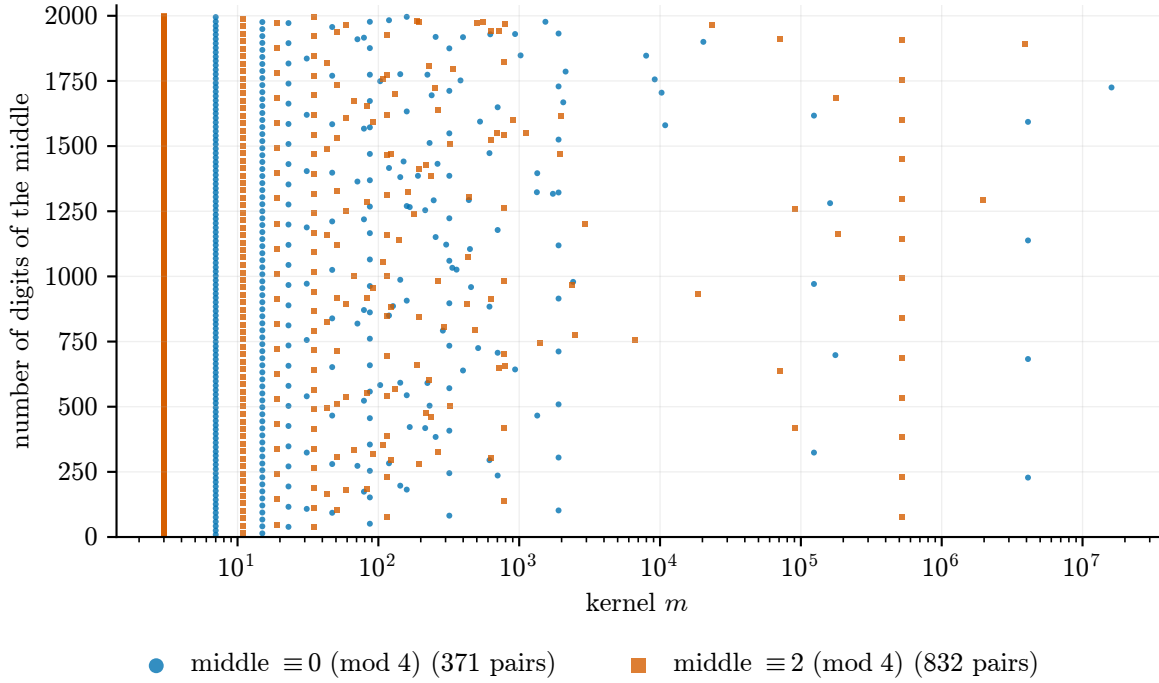


Figure 3: **The pairs at distance two stand in towers.** The pairs at distance two with middle below 10^{2000} and kernel at most 10^8 , each placed at its kernel m and the number of digits of its middle. The pairs of one kernel stand in a column at equal distances, the towers of Proposition II.6.2. The kernel $m = 3$ alone gives 583 of them. The column at $m = 525\,003$, with $m' = 3$, is the kernel from the discussion of Question II.6.7.

6 Pairs at distance two

A *pair at distance two* is a pair $n - 1, n + 1$ of powerful numbers, and n is its *middle*. By Lemma I.1.4 in [Part I] the middle is even, and $n = T_k(m)$ for exactly one squarefree odd $m \geq 3$ with $T_1(m)$ even and one odd multiple k of m' . The kernel satisfies $m \equiv 7 \pmod{8}$ when $4 \mid n$, and $m \equiv 3 \pmod{8}$ when $n \equiv 2 \pmod{4}$. We write

$$P_2(N) = \#\{n \leq N : n - 1 \text{ and } n + 1 \text{ are powerful}\},$$

and $P_2(N; M)$ for the same count restricted to kernels $m \leq M$. Proposition I.6.2 in [Part I] lists the middles divisible by four in a box. The next statement does the same for the other middles, and the rest of the section counts both. Figure 3 shows all these pairs at once.

6.1 The middles that are not divisible by four

Proposition II.6.1 (the middles that are not divisible by four). [computed]

- (a) Among the 10132121 squarefree $m \equiv 3 \pmod{8}$ with $m \leq 10^8$, exactly 86 pass the discard criterion of Theorem I.3.1 in [Part I] for the height 10^{2000} ; the largest of them

is 3 926 299. Their lattice points (m, k) with $T_k(m) < 10^{2000}$ number 883. For 51 of them $T_1(m)$ is odd, and the other 832 are exactly the middles $n \equiv 2 \pmod{4}$ of pairs at distance two with $n < 10^{2000}$ and $\text{sqfree}(n^2 - 1) \leq 10^8$.

(b) These 832 middles have 57 different kernels, and the kernel $m = 3$ alone gives 583 of them.

(c) The list in (a) contains every middle $n \equiv 2 \pmod{4}$ with $n < 1.83 \cdot 10^{18}$, every such middle with $n < 3.88 \cdot 10^{20}$ if Reinhart's extended search missed nothing (Remark I.2.5 in [Part I]), and every such middle with $n < 10^{36}/27$ whose kernel does not divide U_1 .

Proof. (a) is the computation of Theorem I.3.1 in [Part I], run with the residue class 3 in place of 7, together with Lemma I.1.4 in [Part I]: a lattice point with T_1 odd has T_k odd and gives no pair. (b) is a count in the list of (a). (c) Let n be such a middle and m its kernel. Remark I.2.2 in [Part I] gives the bound $T_k \geq (m^{3/2}/m')^{m'}$ in this class too, so the proof of Corollary I.2.3 in [Part I] applies: if $m \nmid U_1$, then $m^{4.5}/27 < n$, and $m < 10^8$ for $n < 10^{36}/27$. If $m \mid U_1$, then $m^{3/2} < T_1(m) \leq n$, so $n < 1.83 \cdot 10^{18}$ forces $m < 1.5 \cdot 10^{12}$, the range in which Reinhart's list of such kernels is complete [11, Remark 5.5], and $n < 3.88 \cdot 10^{20}$ forces $m < 5.325 \cdot 10^{13}$, the range of his extended search [12]. The list has two kernels $m \equiv 3 \pmod{8}$, 1 752 299 below the smaller bound and 20 256 129 307 923 below the larger one. The first lies in the box of (a); its T_1 has 117 digits and is odd. For the second, T_1 is even and has 99 134 digits, so its first middle is far above $3.88 \cdot 10^{20}$. $\square$

In words. The middles of the form $4j + 2$ come from the kernels m of the form $8j + 3$. In the box of kernels up to 10^8 and middles below 10^{2000} there are 832 of them, and most come from a single kernel, $m = 3$. None of them is the middle of a triple, since such a middle is divisible by 2 exactly once.

6.2 The count is a sum over towers

For a kernel m with $T_1(m)$ even, the middles $T_k(m)$ with k an odd multiple of m' form the tower over m .

Proposition II.6.2 (the count is a sum over towers). [proved] *For all integers $N, M \geq 1$,*

$$P_2(N; M) = \sum_{\substack{3 \leq m \leq M \\ \text{odd, squarefree} \\ T_1(m) \text{ even}}} \# \{ k \text{ an odd multiple of } m' : k \log \varepsilon_m \leq \log(2N) \}.$$

Proof. By Lemma I.1.4 in [Part I] the middles $n \leq N$ with kernel $m \leq M$ correspond one to one to the pairs (m, k) in the sum with $T_k(m) \leq N$. It remains to show that $T_k \leq N$ exactly when $\varepsilon_m^k \leq 2N$. We have $2T_k = \varepsilon_m^k + \varepsilon_m^{-k}$ with $0 < \varepsilon_m^{-k} < 1$. If $T_k \leq N$, then $\varepsilon_m^k < 2T_k \leq 2N$. If $\varepsilon_m^k \leq 2N$, then $2T_k < 2N + 1$, and since $2T_k$ is an integer, $2T_k \leq 2N$. $\square$

In words. Each kernel carries a tower of pairs, one on every rung k , and the rungs are evenly spaced on a logarithmic scale: from one rung to the next the middle grows by the factor $\varepsilon_m^{2m'}$. Counting the pairs below a height is counting rungs below that height, tower by tower.

We evaluated the right side for the kernels that pass the discard criterion in both classes and compared it with the lattice points that the searches of Theorem I.3.1 in [Part I] and Proposition II.6.1 found. Table 6 shows six heights; the two agree at every height in both classes. The last two columns are a negative control: the same count with the step $\log T_1$ in place of $\log \varepsilon_m$, an error of an early version of our program, gives larger counts, which are wrong.

h	pairs, counted and by the tower sum		negative control (step $\log T_1$)	
	middle $\equiv 0 \pmod{4}$	middle $\equiv 2 \pmod{4}$	middle $\equiv 0 \pmod{4}$	middle $\equiv 2 \pmod{4}$
100	14	37	21	67
250	39	97	53	168
500	83	199	111	344
1000	176	410	224	695
1500	270	616	345	1049
2000	371	832	467	1404

Table 6: Pairs at distance two with middle at most 10^h and kernel at most 10^8 . The counted pairs and the tower sum of Proposition II.6.2 agree in every row.

Each tower adds one pair every time $\log_{10} N$ grows by $2m' \log_{10} \varepsilon_m$. So $P_2(10^h; M)$ grows with h at the average rate

$$c_M = \sum_{\substack{3 \leq m \leq M \\ T_1(m) \text{ even}}} w(m), \quad w(m) = \frac{1}{2m' \log_{10} \varepsilon_m},$$

pairs per decimal digit. For the towers of Table 6 the rate is 0.1817 for the middles divisible by four and 0.4134 for the others. This rate belongs to the kernels up to M ; it is not a rate for P_2 itself, because kernels above M add towers of their own. Whether c_M stays bounded as M grows is Question II.6.7.

Corollary II.6.3 (lower bound at every height). [computed] *For every integer $h \geq 1$,*

$$P_2(10^h) \geq 0.595 h - 122.$$

For the middles divisible by four alone, the count is at least $0.1817 h - 60$.

Proof. Take the 122 kernels $m \leq 10^8$ with $T_1(m)$ even that pass the discard criterion, 60 of them with $m \equiv 7 \pmod{8}$ and 62 with $m \equiv 3 \pmod{8}$. By Proposition II.6.2 the tower over m has a pair with middle at most 10^h for every odd j with $j \leq y_m$, where $y_m = (h + \log_{10} 2)/(m' \log_{10} \varepsilon_m)$. There are $\lfloor (y_m + 1)/2 \rfloor > y_m/2 - 1/2$ such j , and $y_m/2 \geq h w(m)$. Different kernels give different middles. Summing over the kernels gives more than $h \sum_m w(m) - 122/2$ pairs. The sum of the weights is at least 0.595, and at least 0.1817 over the first class, which gives both bounds with half the subtracted constants; the statement keeps the larger ones. $\square$

In words. Every tower keeps producing pairs forever, one per rung. Adding up the rungs of the towers we know gives a lower bound for the number of pairs below any height, however large. Any single tower already gives a bound of this shape; the constant here is the sum

over the 122 towers that pass the discard criterion. For heights below 10^{2000} the count over these towers is exactly the number of pairs with kernel at most 10^8 , by Proposition II.6.2.

That there are infinitely many pairs at distance two is classical [14, p. 272]; Corollary II.6.3 only puts a number on how fast they appear.

6.3 Where the pairs live

For an odd $J \geq 3$ let

$$E_J(N) = \#\{m \text{ odd, squarefree} : m' < J, T_1(m) \text{ even}, T_{m'}(m) \leq N\},$$

the kernels whose first rung is a middle at most N and whose index m' is below J . For $J = 3$ these are the kernels with $m \mid U_1$ and $T_1(m)$ even; those with $m \equiv 7 \pmod{8}$ form wing (ii) of Part I (Corollary I.2.4 in [Part I]).

Proposition II.6.4 (localization of the pair count). [proved] *For every integer $N \geq 2$ and every odd $J \geq 3$,*

$$P_2(N) \leq 6J (J^J N)^{2/(3J)} + \frac{2}{\log 3} (J^J N)^{1/(3J)} \log N + \frac{2 \log N}{\log 3} E_J(N), \quad E_J(N) \leq P_2(N).$$

In particular $P_2(N) \leq 18 (27N)^{2/9} + \frac{2}{\log 3} (27N)^{1/9} \log N + \frac{2 \log N}{\log 3} E_3(N)$. Moreover,

$$E_3(N) \leq P_2(N) \leq E_3(N) + 41 N^{2/9} + 3 N^{1/9} \log N \leq E_3(N) + 51 N^{2/9}.$$

Proof. By Lemma I.1.4 in [Part I] every middle $n \leq N$ is $T_k(m)$ for one kernel m and one odd multiple k of m' , with $T_1(m)$ even. For a fixed kernel, $T_k \geq T_1^k$ and $T_1 > \sqrt{m}$, so there are fewer than $2 \log N / \log m$ indices k with $T_k \leq N$.

Kernels with $m' \geq J$. By Remark I.2.2 in [Part I], $N \geq T_k \geq (m^{3/2}/m')^{m'}$. The function $f(x) = x(1.5 \log m - \log x)$ increases on $[3, m]$ for $m \geq 8$, as in the proof of Corollary I.2.3 in [Part I], and for $m = 7$ one checks $f(3) < f(5) < f(7)$ directly; for $m = 3$ we have $m' = 3 = J$, and $m = 5$ does not occur, because $m \equiv 3$ or $7 \pmod{8}$. So $N \geq (m^{3/2}/J)^J$, that is, $m \leq M_J := (J^J N)^{2/(3J)}$. Using $\sum_{3 \leq m \leq M} 1/\log m \leq \sqrt{M}/\log 3 + 2M/\log M$ and $\log M_J \geq (2/(3J)) \log N$, these kernels contribute at most

$$2 \log N (2M_J/\log M_J + \sqrt{M_J}/\log 3) \leq 6JM_J + (2/\log 3) \sqrt{M_J} \log N$$

middles.

Kernels with $m' < J$. Then $T_{m'}(m) \leq T_k(m) \leq N$, and $T_{m'}$ is even because T_1 is even and m' is odd. So m is counted by $E_J(N)$, and it contributes fewer than $2 \log N / \log 3$ middles.

Finally, $m \mapsto T_{m'}(m)$ maps the kernels counted by $E_J(N)$ to different middles at most N ; each $T_{m'}(m)$ is a middle by the converse direction of Lemma I.1.4 in [Part I]. So $E_J(N) \leq P_2(N)$.

For the last statement take $J = 3$. The kernels with $m' \geq 3$ contribute at most $18(27N)^{2/9} + (2/\log 3)(27N)^{1/9} \log N$ middles, as above. A kernel with $m' < 3$ has $m' = 1$, because m' is odd, so $m \mid U_1$; its middles are the T_k with k odd, and the first of them, T_1 , is counted by $E_3(N)$. A second middle $T_k \leq N$ with $k \geq 3$ requires $T_1^3 \leq T_3 \leq N$. Since $m \mid U_1$, we have $U_1 \geq m$ and $T_1^2 = 1 + mU_1^2 > m^3$, so $m < N^{2/9}$, and the number of odd $k \geq 3$

with $T_k \leq N$ is at most $\log N/(2 \log T_1) < \log N/(3 \log m)$. With the bound for $\sum 1/\log m$ above and $M = N^{2/9}$, these kernels contribute at most $3N^{2/9} + N^{1/9} \log N/(3 \log 3)$ further middles. Adding up, with $18 \cdot 27^{2/9} + 3 < 41$ and $(2/\log 3) 27^{1/9} + 1/(3 \log 3) < 3$, gives the first bound, and $N^{1/9} \log N \leq (9/e) N^{2/9}$ gives the second. $\square$

In words. There are two kinds of kernels. Those with a large index m' must be small, because their towers start high; together they give only at most of the order $N^{2/(3J)}$ pairs. All other pairs come from the kernels whose index is small, whose number is the open point, and each of these gives only a logarithmic number of pairs. So the question of how many pairs there are is a question about how many kernels with small index there are. A finer count shows that a kernel with $m \mid U_1$ almost always gives only its first pair: a second one needs $T_1^3 \leq N$, which only kernels below $N^{2/9}$ can afford. So the pairs and these kernels agree up to $N^{2/9}$.

Corollary II.6.5 (what the localization says, and what it does not say). [proved]

- (a) For every fixed odd $J \geq 3$, all but $O_J(N^{2/(3J)})$ of the middles counted by $P_2(N)$ have a kernel with $m' < J$.
- (b) Since $E_J(N) \leq P_2(N)$, Proposition II.6.4 gives no bound for $P_2(N)$ by itself. Conversely, every bound $E_J(N) \ll N^\theta$ gives $P_2(N) \ll N^{\max(2/(3J), \theta)} \log N$. For $J = 3$ the factor $\log N$ is not needed: every bound $E_3(N) \ll N^\theta$ gives $P_2(N) \ll N^{\max(2/9, \theta)}$, and for $\theta \geq 2/9$ the two bounds are equivalent.

Proof. Both parts are read off from Proposition II.6.4. $\square$

In words. The inequality moves the problem, it does not solve it: counting pairs is as hard as counting the kernels with small index, and for those we have no better bound than the one for all pairs.

Remark II.6.6 (where Tao's argument is weakest). [proved] For distance two, Chan [5, Theorem 3] proves $P_2(N) \ll N^{2/5}(\log N)^2$, Tao [16, Corollary 2.11] proves $P_2(N) \ll N^{2/5+o(1)}$ within a more general statement, and Blomer and Schöbel [3, Theorem 3] prove $P_2(N) \ll N^\gamma$ for every $\gamma > 61/180$, for every fixed nonzero difference. Tao [16, Remark 2.12] notes that his argument is weakest when $n - 1 = n_1^2 n_2^3$ and $n + 1 = m_1^2 m_2^3$ with n_1, n_2, m_1, m_2 all of size about $N^{1/5}$. If moreover n_2 and m_2 are squarefree, the kernel is $m = n_2 m_2$, of size about $N^{2/5}$. A kernel with $m \nmid U_1$ would give $n > m^{4.5}/27$ by Corollary I.2.3 in [Part I] (with Remark I.2.2 in [Part I] when $n \equiv 2 \pmod{4}$), which exceeds N for large N ; so $m \mid U_1$, and $k = 1$, because $k \geq 3$ would give $n \geq T_1^3 > m^{9/2}$. Precisely, if $n \leq N$ and $m \geq (27N)^{2/9}$, then $m \mid U_1(m)$ and $k = 1$. So, under the assumption that n_2 and m_2 are squarefree, the case in which Tao's argument is weakest lies in the set counted by $E_3(N)$, the first rungs of kernels that divide U_1 . We do not claim the converse: E_3 also counts kernels of other sizes.

In words. The bounds cited above come from counting solutions of one equation. Tao says where his counting is weakest, and that place lies inside the kernels of the localization when the numbers n_2 and m_2 are squarefree.

6.4 How fast does the count grow?

Question II.6.7 (growth of the tower rate). [open] Does $c_M = \sum_{m \leq M} w(m)$, the sum over odd squarefree m with $T_1(m)$ even of $w(m) = 1/(2m' \log_{10} \varepsilon_m)$, stay bounded as $M \rightarrow \infty$?

The partial sums increase with M , so either they converge or they tend to infinity. We separate what is proved, what is measured and what is heuristic.

Proved and computed. Proposition II.6.2 writes $P_2(N; M)$ as an exact tower sum, and Corollary II.6.3 gives $P_2(10^h) \geq 0.595h - 122$. Neither answer to the question is excluded. An answer would not give an asymptotic formula for $P_2(N)$ by itself: that also needs to know how many kernels have a tower that reaches below N , and this is governed by the same unknown kernels.

Measured. Table 7 gives c_M for $M = 10, 10^2, \dots, 10^7$ in both classes. In both, the increase from one decade to the next becomes smaller. Single kernels matter: in the second class the kernel $m = 525\,003$, with $m' = 3$, alone gives about 45 % of the increase from $M = 10^5$ to $M = 10^6$.

$\log_{10} M$	$c_M, m \equiv 7 \pmod{8}$	$c_M, m \equiv 3 \pmod{8}$
1	0.0594	0.2914
2	0.1341	0.3666
3	0.1711	0.4027
4	0.1936	0.4212
5	0.2047	0.4366
6	0.2151	0.4513
7	0.2217	0.4566

Table 7: The partial sums c_M of Question II.6.7, over all kernels up to M with $T_1(m)$ even, without a height limit.

Heuristic, not a claim. The quotient m/m' is the product of the primes of m that divide U_1 . A simple model says that a prime $p \mid m$ divides U_1 with probability $1/p$. For the kernels up to 10^7 with T_1 even and the primes $p \leq 31$, the measured frequency divided by $1/p$ lies between 0.94 and 1.01 in the first class and between 0.91 and 1.04 in the second; $m' = m$ holds for 87.1 % and 86.3 % of the kernels. Under the model, with independence over the primes of m , and if $\log \varepsilon_m$ is of the order $\sqrt{m}$, the expected contribution of a kernel to c_M is about $m^{-3/2+o(1)}$, and the expected sum converges. The model is checked only for $p \leq 31$, and $\log \varepsilon_m$ is far below $\sqrt{m}$ in families such as $m = t^2 - 2$ (Remark I.8.3 in [Part I]). The sum could still diverge through kernels with a small index m' ; whether they occur often enough is exactly what the question asks. The example above shows why more decades of the same computation would say little: a single such kernel can supply a large part of a decade.

In words. Every kernel adds a small amount to the rate at which pairs appear. The question is whether these amounts add up to a finite total. Our numbers grow more and more slowly, which is compatible with a finite total and equally with a slowly growing one; one kernel with a small index can change the picture, so the data do not decide it.

7 Distance one along the Pell route

Section 2 describes a pair $a, a+1$ by the two squarefree parts b and d . This section uses their product instead. That puts every pair on the equation $x^2 - Dy^2 = 1$ for one kernel D , as Part I does for the middles at distance two, and it makes the question of Part I easy to ask: a triple is a pair whose third neighbor is powerful as well.

Proposition II.7.1 (pairs on one Pell equation). [proved] *Let $a \geq 1$, $T = 2a + 1$ and $D = \text{sqfree}(a(a+1))$, and write $a(a+1) = DR^2$. Then $T^2 - D(2R)^2 = 1$, and a and $a+1$ are both powerful if and only if $D \mid R$.*

Proof. We have $T^2 - 1 = 4a(a+1) = 4DR^2$. Since a and $a+1$ are coprime, every prime factor of $a(a+1)$ divides exactly one of them, with the same exponent; so both are powerful if and only if $a(a+1)$ is. By Lemma II.2.1, $a(a+1) = DR^2$ is powerful if and only if $D \mid R$. $\square$

In words. Two neighbors are both powerful exactly when their product is. The product is a squarefree number D times a square R^2 , and it is powerful exactly when D divides R . So the pairs with a given D are found among the solutions of one Pell equation.

For a family (b, d) of Section 2 one has $D = bd$ and $R = XY$, so Proposition II.7.1 is Proposition II.2.2 written with one kernel instead of two. The solutions of $x^2 - Dy^2 = 1$ with x odd carry these pairs; the solutions with x even carry the middles at distance two of Lemma I.1.4 in [Part I]. A triple of consecutive powerful numbers is a pair $a, a+1$ for which $a-1$ or $a+2$ is powerful as well, and one of the two is excluded at once: the even member of the pair is powerful, hence divisible by 4, so its neighbor at distance two is 2 more than a multiple of 4 and not powerful. We call the other one the *third number* of the pair.

Proposition II.7.2 (the third number). [computed]

- (a) *For the 12159 squarefree $D \leq 20\,000$, the pairs $a, a+1$ of Proposition II.7.1 with $2a+1 < 10^{2000}$ number 6056. The program ran each kernel one rung further, which adds the next pair above this bound for 7 kernels; together these are the 6063 pairs of (b) and (c). A direct search for consecutive powerful numbers up to 10^{12} , which does not use the equation, finds 18 pairs, and the route with $D \leq 10^5$ finds the same 18.*
- (b) *For 4991 of these pairs the third number has a prime factor $p < 5 \cdot 10^5$ with exponent 1, and for 154 more a prime factor $p < 5 \cdot 10^6$ with exponent 1. For the remaining 918, all prime factors below $5 \cdot 10^6$ occur at least twice. After removing them, the rest of the third number is a prime in 24 cases, 5 of them proved by a deterministic test [15] and the other 19, with up to 1687 digits, by elliptic curve certificates [7, 1] that PARI/GP [17] produced and a separate program of ours checked; in 15 further cases the elliptic curve method finds a prime factor with exponent 1.*
- (c) *So for all but 879 of the 6063 pairs a prime with exponent 1 is known for the third number, and it is proved prime. For the 879 others no witness is known; the smallest of their third numbers has 81 digits.*

The computation. (a) For each D we solved $x^2 - Dy^2 = 1$, took the powers $T_k + U_k\sqrt{D}$ with T_k odd below 10^{2000} , and kept those with $D \mid U_k$ and $T_k \equiv \pm 1 \pmod{8}$. For odd D the

second condition follows from the first; for $D = 2D_0$ the two together say $2D_0 \mid U_k/2$. In both cases they are equivalent to $D \mid U_k/2 = R$, so by Proposition II.7.1 these are exactly the pairs. The comparison in (a) was made up to 10^{12} with a separate program that lists all powerful numbers. (b) The third number is $a - 1 = (T - 3)/2$ or $a + 2 = (T + 3)/2$. For an odd prime p , the exponent of p in it is the exponent of p in $T \mp 3$, which we computed from T_k modulo p^2 . For the remaining pairs we computed the third number exactly, divided out all primes below $5 \cdot 10^6$ and tested the rest; a prime rest divides the number exactly once. As controls, the program finds for 5 pairs of the first group the prime found there, and for the pair with $(D, k) = (3, 24)$ it finds that the third number $a - 1$ is itself a prime. $\square$

In words. For every pair we tried to show that its third neighbor is not powerful, by finding a prime that divides it only once. This worked for all but 879 pairs. For those, we could not decide it; that does not make them triples. A triple among them with $2a + 1 < 10^{2000}$ would also have to avoid Part I: its middle is below 10^{2000} , so by Theorem I.3.1 in [Part I] its kernel in the sense of Part I would exceed 10^8 .

8 Open questions

The problems left open in this part are the following. Each is stated where it arises; we collect them here.

- (1) Erdős asked whether the number $N_1(x)$ of pairs $n, n + 1$ of powerful numbers with $n + 1 \leq x$ is bounded by a power of $\log x$ [4]. Proposition II.4.1 gives $N_1(x) \gg \log x$ with an explicit constant; nothing in this part bounds $N_1(x)$ from above.
- (2) Question II.5.3: does the sum of $1/\log F$ over all families converge?
- (3) Question II.6.7: does the rate c_M of the towers at distance two stay bounded?
- (4) The question below, which Corollary II.6.5 and Remark II.6.6 single out.

Question II.8.1 (the kernels that divide U_1). [open] Is there a $\theta < 61/180$ with $E_3(N) \ll N^\theta$? Since $E_3(N) \leq P_2(N)$, Blomer and Schöbel give $E_3(N) \ll N^\gamma$ for every $\gamma > 61/180$ [3, Theorem 3]; the question is whether the squarefree odd m with $m \mid U_1(m)$, $T_1(m)$ even and $T_1(m) \leq N$ are rarer than that.

By Corollary II.6.5(b), a positive answer gives $P_2(N) \ll N^{\max(2/9, \theta)}$, and for $\theta \geq 2/9$ it is equivalent to that bound. Either way it would improve the best exponent we know for distance two. Reuss's exponent $29/100$ is for distance one [13, Theorem 4], and for distance two the best exponent we know is the $61/180$ of Blomer and Schöbel. Remark II.6.6 shows that the case in which Tao's argument is weakest lies in E_3 when n_2 and m_2 are squarefree. What is known is small: the known kernels counted by E_3 are the members of Reinhart's list with $T_1(m)$ even. Their number below $1.5 \cdot 10^{12}$ is 1 [11, Remark 5.5], and below $5.325 \cdot 10^{13}$ it is 3 on his extended list [12], if that list is complete (Remark I.2.5 in [Part I]); each of them has $T_1(m) > 10^{94}$ (Proposition I.6.3 in [Part I] for the class 7, and 99 134 digits for the kernel of class 3), and every other kernel has $T_1(m) > m^{3/2}$ by Corollary I.2.3 in [Part I]. So $E_3(N) = 0$ for $N < 1.83 \cdot 10^{18}$, and for $N < 3.88 \cdot 10^{20}$ under that assumption. A bound for E_3 is a statement about fundamental units, of the kind that Part I also needs in its wing (ii).

In words. Four questions remain. Two ask whether rates that we computed add up to a finite total. One is Erdős’s question itself. The last asks how many kernels of the kind that divides U_1 there are; a positive answer would improve the best bound we know for pairs at distance two, and it is the same gap that Part I runs into.

The type of every statement

Every statement of this part carries one of six labels, printed after its number. The table lists them; it is generated from the text by the build and checked against a second, independent reading of the source.

statement	title	type
Lemma II.2.1	powerful numbers	proved
Proposition II.2.2	Walker [20, Sections 2 and 3]	literature
Proposition II.2.3	Walker [20, Theorems 2.2, 2.5, 3.2 and 3.5]	literature
Proposition II.3.1	the known pairs	computed
Proposition II.3.2	completeness inside the families	computed
Proposition II.4.1	explicit lower bound	computed
Definition II.5.1	comparison function; a heuristic	heuristic
Proposition II.5.2	the families with a square overtake the comparison function	computed
Question II.5.3	convergence of the rates	open
Proposition II.6.1	the middles that are not divisible by four	computed
Proposition II.6.2	the count is a sum over towers	proved
Corollary II.6.3	lower bound at every height	computed
Proposition II.6.4	localization of the pair count	proved
Corollary II.6.5	what the localization says, and what it does not say	proved
Remark II.6.6	where Tao’s argument is weakest	proved
Question II.6.7	growth of the tower rate	open
Proposition II.7.1	pairs on one Pell equation	proved
Proposition II.7.2	the third number	computed
Question II.8.1	the kernels that divide U_1	open

The types: *proved*, proved in this part; the proof may use published results (6); *literature*, due to the cited source, sometimes with a proof given here for completeness (2); *computed*, the result of an exact computation, or a proof that uses one of our computations (7); *heuristic*, a heuristic, used as a yardstick only (1); *open*, an open question (3).

Data and code availability

Every number of this part that we computed is generated from a result file of a data deposit or from a cited constant; no computed number is typed by hand. The data deposit is archived with the collected volume, DOI 10.5281/zenodo.23127547.

What the deposit contains. The deposit has two parts. The core holds what the text cites: (i) the result files, in JSON and plain text, behind every computed number, together

with the raw output of the scripts that wrote them; (ii) the scripts, written in Python, each stating before it runs what it expects to find and the positive and negative controls that it runs; (iii) a manifest with the SHA-256 hash and the size of every file, which the build checks before it reads a result file; (iv) the decimal expansions of the building blocks of Part I that the searches left open (Proposition I.6.8 in [Part I]), including those that the algebraic split of Remark I.6.13 in [Part I] settles; (v) the certificates and the second routes for the computations that settle a building block; and (vi) the file `OPEN_PROBLEMS.md`, which lists the open problems of the volume. The archive holds the remaining scripts and results of the investigation, including negative results that the text does not print.

Where the files are. The collected volume and its three parts are published as four linked records on Zenodo, and the same files are in one repository on GitHub. The record of the collected volume (DOI 10.5281/zenodo.23127547) holds the volume and the whole deposit, the core and the archive described above. The record of each part holds that part and the result files that its text cites, and links to the record of the volume. The repository (github.com/berkay-yueksel-sayim/erdos-364-365-powerful-numbers-pell) holds the four PDFs and the deposit, and its README lists the four records. The records of Parts I, II and III have the DOIs 10.5281/zenodo.23127703, 10.5281/zenodo.23127705 and 10.5281/zenodo.23127707.

How the numbers get into the text. The source of the text refers to each computed number by a key. A build tool reads the result file named for the key, verifies it against the manifest, computes the number in its printed form, and writes it into the text; it stops if a key has no source or if a file differs from the manifest. The same tool generates the table of the types of the statements and the list of open problems from the source of the text, and it is part of the core.

Software. The scripts use Python 3.12 with gmpy2 2.3, SymPy 1.14 and NumPy 2.4. The elliptic curve method and its relatives are run with GMP-ECM 7.0.6 [21], used as an installed program inside a container image built from the Debian package and not part of the deposit; the recipe of the image is in the core. The elliptic curve certificates for primality were produced in the same way with PARI/GP 2.17.2 [17] and are part of the deposit; a separate program of ours checks each of them, so the result does not depend on that program. Further work is planned to support these certificates with a second prover written independently by us. The text is set with pdf_TE_X.

Licenses. The text and the data are licensed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), and the code under the Apache License, Version 2.0. Programs of others that the scripts use as tools, such as GMP-ECM, are not part of the deposit and keep their own licenses.

Use of AI tools

Generative AI tools were used throughout this work: large language models of the Claude family by Anthropic, mainly Claude Opus 5 and 5.5, and also Claude Fable 5.1 and Claude

Sonnet 5.5 and 5, between August and October 2026, through the Claude Code environment. They were used to draft the text and the mathematical arguments, to write the computer programs, to run and check the computations, and to search for and read the literature. The author works independently, is not trained as a number theorist, and learned the subject actively during the work; the tools made it possible to carry out and check computations on this scale. The author led the work in stages: each stage was set by a prompt that stated what it should achieve, within a stage the tools followed a plan agreed with the author, and the author considered the results before giving the next prompt, revising the plan when the results required it. The author contributed ideas where they helped and framed how a problem should be attacked, decided which statements to keep, and is responsible for the content. The programs of the project, including its own checking and build tools, were developed with the AI tools, and the author started the long computations, such as the factor searches, on the author's own machine.

Several parts of the checking do not rest on these tools. The elliptic curve primality certificates were produced by PARI/GP 2.17.2 and are verified by a separate program, and the factor searches use GMP-ECM 7.0.6. Every computed number is generated by a program from deposited data or from a cited constant, and the programs carry positive and negative controls. Proofs, numbers and citations were also checked in independent passes by further instances of the AI tools that were given no access to the drafting context. These tools are not authors.

References

- [1] A. O. L. Atkin and F. Morain, *Elliptic curves and primality proving*, Math. Comp. **61** (1993), no. 203, 29–68.
- [2] Paul T. Bateman and Emil Grosswald, *On a theorem of Erdős and Szekeres*, Illinois J. Math. **2** (1958), no. 1, 88–98.
- [3] Valentin Blomer and Anita Schöbel, *Twins of powerful numbers*, Funct. Approx. Comment. Math. **49** (2013), no. 2, 349–356.
- [4] T. F. Bloom, *Erdős Problem #365*, <https://www.erdosproblems.com/365>, Accessed 2026-09-28.
- [5] Tsz Ho Chan, *Twin squareful numbers*, J. Aust. Math. Soc. **93** (2012), no. 1-2, 43–51.
- [6] P. Erdős and G. Szekeres, *Über die Anzahl der Abelschen Gruppen gegebener Ordnung und über ein verwandtes zahlentheoretisches Problem*, Acta Litt. Sci. Szeged **7** (1934), 95–102.
- [7] S. Goldwasser and J. Kilian, *Almost all primes can be quickly certified*, Proceedings of the Eighteenth Annual ACM Symposium on Theory of Computing (STOC '86), ACM, 1986, pp. 316–329.
- [8] S. W. Golomb, *Powerful numbers*, Amer. Math. Monthly **77** (1970), no. 8, 848–852.
- [9] Ibrahim Mian and Shayaan Siddique, *A kernel-certified verification of the Erdős–Mollin–Walsh conjecture below 10^{14}* , 2026, arXiv:2609.25011 (v1).

- [10] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A060355>, 2026, Sequence A060355; b-file by Donovan Johnson (terms $< 10^{22}$).
- [11] Andreas Reinhart, *On orders in quadratic number fields with unusual sets of distances*, Acta Arith. **211** (2023), 61–92, arXiv:2305.09267.
- [12] ———, *A counterexample to the Pellian equation conjecture of Mordell*, 2024, arXiv:2402.09827 (v2).
- [13] T. Reuss, *Pairs of k -free numbers, consecutive square-full numbers*, 2014, arXiv:1212.3150v2.
- [14] W. A. Sentance, *Occurrences of consecutive odd powerful numbers*, Amer. Math. Monthly **88** (1981), no. 4, 272–274.
- [15] Jonathan Sorenson and Jonathan Webster, *Strong pseudoprimes to twelve prime bases*, Math. Comp. **86** (2017), no. 304, 985–1003.
- [16] Terence Tao, *Products of consecutive integers with unusual anatomy*, 2026, arXiv:2603.27990 (v2).
- [17] The PARI Group, *PARI/GP, version 2.17.2*, Univ. Bordeaux; Debian package pari-gp 2.17.2-1.
- [18] Wouter van Doorn, *Three-term arithmetic progressions of consecutive powerful numbers*, 2026, arXiv:2605.06697 (v1).
- [19] D. T. Walker, *On the Diophantine equation $mX^2 - nY^2 = \pm 1$* , Amer. Math. Monthly **74** (1967), no. 5, 504–513.
- [20] David T. Walker, *Consecutive integer pairs of powerful numbers and related Diophantine equations*, Fibonacci Quart. **14** (1976), no. 2, 111–116.
- [21] Paul Zimmermann et al., *GMP-ECM, version 7.0.6*, The GMP-ECM Project Team; Debian package gmp-ecm 7.0.6+ds-1.